

Extracted Code from Jupyter Notebook

Code Cell 1

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
```

Code Cell 2

```
df = pd.read_csv('/content/drive/MyDrive/OnlineRetail.csv', encoding='latin1')

print(df.head())
print(df.shape)
print(df.info())
```

Code Cell 3

```
df = df[['Quantity', 'UnitPrice', 'Country']]

print(df.head())
```

Code Cell 4

```
df.dropna(inplace=True)

print(df.isnull().sum())
```

Code Cell 5

```
le = LabelEncoder()

df['Country'] = le.fit_transform(df['Country'])

print(df.head())
```

Code Cell 6

```
median_quantity = df['Quantity'].median()

df['PurchaseType'] = np.where(
    df['Quantity'] > median_quantity,
    1,
    0
)

print(df.head())
```

Code Cell 7

```
plt.figure(figsize=(6,4))
```

```
df['PurchaseType'].value_counts().plot(kind='bar')

plt.title("Customer Distribution")
plt.xlabel("Purchase Type")
plt.ylabel("Count")

plt.show()
```

Code Cell 8

```
x = df[['Quantity', 'UnitPrice', 'Country']]
y = df['PurchaseType']
```

Code Cell 9

```
x_train, x_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.20,
    random_state=42
)

print("Training Size:", x_train.shape)
print("Testing Size:", x_test.shape)
```

Code Cell 10

```
lr = LogisticRegression()

lr.fit(x_train, y_train)

y_pred_lr = lr.predict(x_test)

acc_lr = accuracy_score(y_test, y_pred_lr)

print("Logistic Regression Accuracy:")
print(acc_lr)
```

Code Cell 11

```
cm_lr = confusion_matrix(y_test, y_pred_lr)

print(cm_lr)
```

Code Cell 12

```
dt = DecisionTreeClassifier(random_state=42)

dt.fit(x_train, y_train)

y_pred_dt = dt.predict(x_test)

acc_dt = accuracy_score(y_test, y_pred_dt)

print("Decision Tree Accuracy:")
print(acc_dt)
```

Code Cell 13

```
cm_dt = confusion_matrix(y_test, y_pred_dt)

print(cm_dt)
```

Code Cell 14

```
knn = KNeighborsClassifier(n_neighbors=5)

knn.fit(X_train,y_train)

y_pred_knn = knn.predict(X_test)

acc_knn = accuracy_score(y_test,y_pred_knn)

print("KNN Accuracy:")
print(acc_knn)
```

Code Cell 15

```
cm_knn = confusion_matrix(y_test,y_pred_knn)

print(cm_knn)
```

Code Cell 16

```
models = [
    'Logistic Regression',
    'Decision Tree',
    'KNN'
]

accuracies = [
    acc_lr,
    acc_dt,
    acc_knn
]

plt.figure(figsize=(8,5))

plt.bar(models,accuracies)

plt.title("Model Accuracy Comparison")

plt.xlabel("Models")

plt.ylabel("Accuracy")

plt.show()
```

Code Cell 17

```
plt.figure(figsize=(8,5))

plt.scatter(df['Quantity'], df['UnitPrice'])

plt.title('Quantity vs Unit Price')
plt.xlabel('Quantity')
plt.ylabel('Unit Price')

plt.show()
```